

7. Analyze what happens when the number of handles in the basic building block for the Alexander horned sphere is doubled, as in the figure at the right.



8. Show that $\mathbb{R}^{2n+1}$ is not a division algebra over $\mathbb{R}$ if $n > 0$ by showing that if it were, then for nonzero $a \in \mathbb{R}^{2n+1}$ the map $S^{2n} \rightarrow S^{2n}$, $x \mapsto ax/|ax|$ would be homotopic to $x \mapsto -ax/|ax|$, but these maps have different degrees.

9. Make the transfer sequence explicit in the case of a trivial covering $\tilde{X} \rightarrow X$, where $\tilde{X} = X \times S^0$.

10. Use the transfer sequence for the covering $S^\infty \rightarrow \mathbb{RP}^\infty$ to compute $H_n(\mathbb{RP}^\infty; \mathbb{Z}_2)$.

11. Use the transfer sequence for the covering $X \times S^\infty \rightarrow X \times \mathbb{RP}^\infty$ to produce isomorphisms $H_n(X \times \mathbb{RP}^\infty; \mathbb{Z}_2) \approx \bigoplus_{i \leq n} H_i(X; \mathbb{Z}_2)$ for all n .

2.C Simplicial Approximation

Many spaces of interest in algebraic topology can be given the structure of simplicial complexes, and early in the history of the subject this structure was exploited as one of the main technical tools. Later, CW complexes largely superseded simplicial complexes in this role, but there are still some occasions when the extra structure of simplicial complexes can be quite useful. This will be illustrated nicely by the proof of the classical Lefschetz fixed point theorem in this section.

One of the good features of simplicial complexes is that arbitrary continuous maps between them can always be deformed to maps that are linear on the simplices of some subdivision of the domain complex. This is the idea of ‘simplicial approximation,’ developed by Brouwer and Alexander before 1920. Here is the relevant definition: If K and L are simplicial complexes, then a map $f: K \rightarrow L$ is **simplicial** if it sends each simplex of K to a simplex of L by a linear map taking vertices to vertices. In barycentric coordinates, a linear map of a simplex $[v_0, \dots, v_n]$ has the form $\sum_i t_i v_i \mapsto \sum_i t_i f(v_i)$. Since a linear map from a simplex to a simplex is uniquely determined by its values on vertices, this means that a simplicial map is uniquely determined by its values on vertices. It is easy to see that a map from the vertices of K to the vertices of L extends to a simplicial map iff it sends the vertices of each simplex of K to the vertices of some simplex of L .

Here is the most basic form of the **Simplicial Approximation Theorem**:

Theorem 2C.1. *If K is a finite simplicial complex and L is an arbitrary simplicial complex, then any map $f: K \rightarrow L$ is homotopic to a map that is simplicial with respect to some iterated barycentric subdivision of K .*

To see that subdivision of K is essential, consider the case of maps $S^n \rightarrow S^n$. With fixed simplicial structures on the domain and range spheres there are only finitely many simplicial maps since there are only finitely many ways to map vertices to vertices. Hence only finitely many degrees are realized by maps that are simplicial with respect to fixed simplicial structures in both the domain and range spheres. This remains true even if the simplicial structure on the range sphere is allowed to vary, since if the range sphere has more vertices than the domain sphere then the map cannot be surjective, hence must have degree zero.

Before proving the simplicial approximation theorem we need some terminology and a lemma. The **star** $\text{St } \sigma$ of a simplex σ in a simplicial complex X is defined to be the subcomplex consisting of all the simplices of X that contain σ . Closely related to this is the **open star** $\text{st } \sigma$, which is the union of the interiors of all simplices containing σ , where the interior of a simplex τ is by definition $\tau - \partial\tau$. Thus $\text{st } \sigma$ is an open set in X whose closure is $\text{St } \sigma$.

Lemma 2C.2. *For vertices $v_1, \dots, v_n$ of a simplicial complex X , the intersection $\text{st } v_1 \cap \dots \cap \text{st } v_n$ is empty unless $v_1, \dots, v_n$ are the vertices of a simplex σ of X , in which case $\text{st } v_1 \cap \dots \cap \text{st } v_n = \text{st } \sigma$.*

Proof: The intersection $\text{st } v_1 \cap \dots \cap \text{st } v_n$ consists of the interiors of all simplices τ whose vertex set contains $\{v_1, \dots, v_n\}$. If $\text{st } v_1 \cap \dots \cap \text{st } v_n$ is nonempty, such a τ exists and contains the simplex $\sigma = [v_1, \dots, v_n] \subset X$. The simplices τ containing $\{v_1, \dots, v_n\}$ are just the simplices containing σ , so $\text{st } v_1 \cap \dots \cap \text{st } v_n = \text{st } \sigma$. $\square$

Proof of 2C.1: Choose a metric on K that restricts to the standard Euclidean metric on each simplex of K . For example, K can be viewed as a subcomplex of a simplex Δ^N whose vertices are all the vertices of K , and we can restrict a standard metric on Δ^N to give a metric on K . Let ε be a Lebesgue number for the open cover $\{f^{-1}(\text{st } w) \mid w \text{ is a vertex of } L\}$ of K . After iterated barycentric subdivision of K we may assume that each simplex has diameter less than $\varepsilon/2$. The closed star of each vertex v of K then has diameter less than ε , hence this closed star maps by f to the open star of some vertex $g(v)$ of L . The resulting map $g: K^0 \rightarrow L^0$ thus satisfies $f(\text{St } v) \subset \text{st } g(v)$ for all vertices v of K .

To see that g extends to a simplicial map $g: K \rightarrow L$, consider the problem of extending g over a simplex $[v_1, \dots, v_n]$ of K . An interior point x of this simplex lies in $\text{st } v_i$ for each i , so $f(x)$ lies in $\text{st } g(v_i)$ for each i , since $f(\text{st } v_i) \subset \text{st } g(v_i)$ by the definition of $g(v_i)$. Thus $\text{st } g(v_1) \cap \dots \cap \text{st } g(v_n) \neq \emptyset$, so $[g(v_1), \dots, g(v_n)]$ is a simplex of L by the lemma, and we can extend g linearly over $[v_1, \dots, v_n]$. Both $f(x)$ and $g(x)$ lie in a single simplex of L since $g(x)$ lies in $[g(v_1), \dots, g(v_n)]$ and $f(x)$ lies in the star of this simplex. So taking the linear path $(1-t)f(x) + tg(x)$, $0 \leq t \leq 1$, in the simplex containing $f(x)$ and $g(x)$ defines a homotopy from f to g . To check continuity of this homotopy it suffices to restrict to the simplex $[v_1, \dots, v_n]$, where

continuity is clear since $f(x)$ varies continuously in the star of $[g(v_1), \dots, g(v_n)]$ and $g(x)$ varies continuously in $[g(v_1), \dots, g(v_n)]$. $\square$

Notice that if f already sends some vertices of K to vertices of L then we may choose g to equal to f on these vertices, and hence the homotopy from f to g will be stationary on these vertices. This is convenient if one is in a situation where one wants maps and homotopies to preserve basepoints.

The proof makes it clear that the simplicial approximation g can be chosen not just homotopic to f but also close to f if we allow subdivisions of L as well as K .

The Lefschetz Fixed Point Theorem

This very classical application of homology is a considerable generalization of the Brouwer fixed point theorem. It is also related to the Euler characteristic formula.

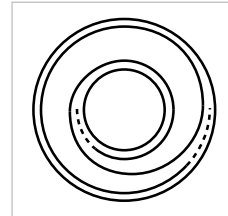
For a homomorphism $\varphi: \mathbb{Z}^n \rightarrow \mathbb{Z}^n$ with matrix $[a_{ij}]$, the trace $\text{tr } \varphi$ is defined to be $\sum_i a_{ii}$, the sum of the diagonal elements of $[a_{ij}]$. Since $\text{tr}([a_{ij}][b_{ij}]) = \text{tr}([b_{ij}][a_{ij}])$, conjugate matrices have the same trace, and it follows that $\text{tr } \varphi$ is independent of the choice of basis for $\mathbb{Z}^n$. For a homomorphism $\varphi: A \rightarrow A$ of a finitely generated abelian group A we can then define $\text{tr } \varphi$ to be the trace of the induced homomorphism $\overline{\varphi}: A/\text{Torsion} \rightarrow A/\text{Torsion}$.

For a map $f: X \rightarrow X$ of a finite CW complex X , or more generally any space whose homology groups are finitely generated and vanish in high dimensions, the **Lefschetz number** $\tau(f)$ is defined to be $\sum_n (-1)^n \text{tr}(f_*: H_n(X) \rightarrow H_n(X))$. In particular, if f is the identity, or is homotopic to the identity, then $\tau(f)$ is the Euler characteristic $\chi(X)$ since the trace of the $n \times n$ identity matrix is n .

Here is the Lefschetz fixed point theorem:

Theorem 2C.3. *If X is a finite simplicial complex, or more generally a retract of a finite simplicial complex, and $f: X \rightarrow X$ is a map with $\tau(f) \neq 0$, then f has a fixed point.*

As we show in Theorem A.7 in the Appendix, every compact, locally contractible space that can be embedded in $\mathbb{R}^n$ for some n is a retract of a finite simplicial complex. This includes compact manifolds and finite CW complexes, for example. The compactness hypothesis is essential, since a translation of $\mathbb{R}$ has $\tau = 1$ but no fixed points. For an example showing that local properties are also significant, let X be the compact subspace of $\mathbb{R}^2$ consisting of two concentric circles together with a copy of $\mathbb{R}$ between them whose two ends spiral in to the two circles, wrapping around them infinitely often, and let $f: X \rightarrow X$ be a homeomorphism translating the copy of $\mathbb{R}$ along itself and rotating the circles, with no fixed points. Since f is homotopic to the identity, we have $\tau(f) = \chi(X)$, which equals 1 since the three path components of X are two circles and a line.



If X has the same homology groups as a point, at least modulo torsion, then the theorem says that every map $X \rightarrow X$ has a fixed point. This holds for example for $\mathbb{R}P^n$ if n is even. The case of projective spaces is interesting because of its connection with linear algebra. An invertible linear transformation $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ takes lines through 0 to lines through 0, hence induces a map $\bar{f}: \mathbb{R}P^{n-1} \rightarrow \mathbb{R}P^{n-1}$. Fixed points of $\bar{f}$ are equivalent to eigenvectors of f . The characteristic polynomial of f has odd degree if n is odd, hence has a real root, so an eigenvector exists in this case. This is in agreement with the observation above that every map $\mathbb{R}P^{2k} \rightarrow \mathbb{R}P^{2k}$ has a fixed point. On the other hand the rotation of $\mathbb{R}^{2k}$ defined by $f(x_1, \dots, x_{2k}) = (x_2, -x_1, x_4, -x_3, \dots, x_{2k}, -x_{2k-1})$ has no eigenvectors and its projectivization $\bar{f}: \mathbb{R}P^{2k-1} \rightarrow \mathbb{R}P^{2k-1}$ has no fixed points.

Similarly, in the complex case an invertible linear transformation $f: \mathbb{C}^n \rightarrow \mathbb{C}^n$ induces $\bar{f}: \mathbb{C}P^{n-1} \rightarrow \mathbb{C}P^{n-1}$, and this always has a fixed point since the characteristic polynomial always has a complex root. Nevertheless, as in the real case there is a map $\mathbb{C}P^{2k-1} \rightarrow \mathbb{C}P^{2k-1}$ without fixed points. Namely, consider $f: \mathbb{C}^{2k} \rightarrow \mathbb{C}^{2k}$ defined by $f(z_1, \dots, z_{2k}) = (\bar{z}_2, -\bar{z}_1, \bar{z}_4, -\bar{z}_3, \dots, \bar{z}_{2k}, -\bar{z}_{2k-1})$. This map is only ‘conjugate-linear’ over $\mathbb{C}$, but this is still good enough to imply that f induces a well-defined map $\bar{f}$ on $\mathbb{C}P^{2k-1}$, and it is easy to check that $\bar{f}$ has no fixed points. The similarity between the real and complex cases persists in the fact that every map $\mathbb{C}P^{2k} \rightarrow \mathbb{C}P^{2k}$ has a fixed point, though to deduce this from the Lefschetz fixed point theorem requires more structure than homology has, so this will be left as an exercise for §3.2, using cup products in cohomology.

One could go further and consider the quaternionic case. The antipodal map of $S^4 = \mathbb{H}P^1$ has no fixed points, but every map $\mathbb{H}P^n \rightarrow \mathbb{H}P^n$ with $n > 1$ does have a fixed point. This is shown in Example 4L.4 using considerably heavier machinery.

Proof of 2C.3: The general case easily reduces to the case of finite simplicial complexes, for suppose $r: K \rightarrow X$ is a retraction of the finite simplicial complex K onto X . For a map $f: X \rightarrow X$, the composition $fr: K \rightarrow X \subset K$ then has exactly the same fixed points as f . Since $r_*: H_n(K) \rightarrow H_n(X)$ is projection onto a direct summand, we clearly have $\text{tr}(f_* r_*) = \text{tr } f_*$, so $\tau(f_* r_*) = \tau(f_*)$.

For X a finite simplicial complex, suppose that $f: X \rightarrow X$ has no fixed points. We claim there is a subdivision L of X , a further subdivision K of L , and a simplicial map $g: K \rightarrow L$ homotopic to f such that $g(\sigma) \cap \sigma = \emptyset$ for each simplex σ of K . To see this, first choose a metric d on X as in the proof of the simplicial approximation theorem. Since f has no fixed points, $d(x, f(x)) > 0$ for all $x \in X$, so by the compactness of X there is an $\varepsilon > 0$ such that $d(x, f(x)) > \varepsilon$ for all x . Choose a subdivision L of X so that the stars of all simplices have diameter less than $\varepsilon/2$. Applying the simplicial approximation theorem, there is a subdivision K of L and a simplicial map $g: K \rightarrow L$ homotopic to f . By construction, g has the property that for each simplex σ of K , $f(\sigma)$ is contained in the star of the simplex $g(\sigma)$. Then $g(\sigma) \cap \sigma = \emptyset$.

for each simplex σ of K since for any choice of $x \in \sigma$ we have $d(x, f(x)) > \varepsilon$, while $g(\sigma)$ lies within distance $\varepsilon/2$ of $f(x)$ and σ lies within distance $\varepsilon/2$ of x , as a consequence of the fact that σ is contained in a simplex of L , K being a subdivision of L .

The Lefschetz numbers $\tau(f)$ and $\tau(g)$ are equal since f and g are homotopic. Since g is simplicial, it takes the n -skeleton K^n of K to the n -skeleton L^n of L , for each n . Since K is a subdivision of L , L^n is contained in K^n , and hence $g(K^n) \subset K^n$ for all n . Thus g induces a chain map of the cellular chain complex $\{H_n(K^n, K^{n-1})\}$ to itself. This can be used to compute $\tau(g)$ according to the formula

$$\tau(g) = \sum_n (-1)^n \operatorname{tr}(g_* : H_n(K^n, K^{n-1}) \rightarrow H_n(K^n, K^{n-1}))$$

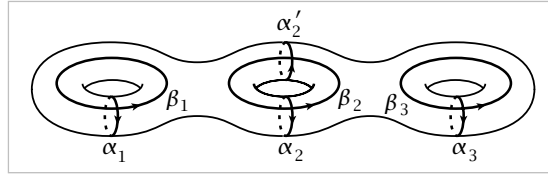
This is the analog of Theorem 2.44 for trace instead of rank, and is proved in precisely the same way, based on the elementary algebraic fact that trace is additive for endomorphisms of short exact sequences: Given a commutative diagram as at the right with exact rows,

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \end{array}$$

then $\operatorname{tr} \beta = \operatorname{tr} \alpha + \operatorname{tr} \gamma$. This algebraic fact can be proved by reducing to the easy case that A , B , and C are free by first factoring out the torsion in B , hence also the torsion in A , then eliminating any remaining torsion in C by replacing A by a larger subgroup $A' \subset B$, with A having finite index in A' . The details of this argument are left to the reader.

Finally, note that $g_* : H_n(K^n, K^{n-1}) \rightarrow H_n(K^n, K^{n-1})$ has trace 0 since the matrix for g_* has zeros down the diagonal, in view of the fact that $g(\sigma) \cap \sigma = \emptyset$ for each n -simplex σ . So $\tau(f) = \tau(g) = 0$. $\square$

Example 2C.4. Let us verify the theorem in an example. Let X be the closed orientable surface of genus 3 as shown in the figure below, with $f : X \rightarrow X$ the 180 degree rotation about a vertical axis passing through the central hole of X . Since f has no fixed points, we should have $\tau(f) = 0$. The induced map $f_* : H_0(X) \rightarrow H_0(X)$ is the identity, as always for a path-connected space, so this contributes 1 to $\tau(f)$. For $H_1(X)$ we saw in Example 2A.2 that the six loops α_i and β_i represent a basis. The map f_* interchanges the homology classes of α_1 and α_3 , and likewise for β_1 and β_3 , while β_2 is sent to itself and α_2 is sent to α'_2 which is homologous to α_2 as we saw in Example 2A.2. So $f_* : H_1(X) \rightarrow H_1(X)$ contributes -2 to $\tau(f)$. It remains to check that $f_* : H_2(X) \rightarrow H_2(X)$ is the identity, which we do by the commutative diagram at the right, where x is a point of X in the central torus and $y = f(x)$. We can see that the



$$\begin{array}{ccc} H_2(X) & \xrightarrow{f_*} & H_2(X) \\ \downarrow \approx & & \downarrow \approx \\ H_2(X, X - \{x\}) & \xrightarrow{f_*} & H_2(X, X - \{y\}) \end{array}$$

left-hand vertical map is an isomorphism by considering the long exact sequence of the triple $(X, X - \{x\}, X^1)$ where X^1 is the 1-skeleton of X in its usual CW structure and x is chosen in $X - X^1$, so that $X - \{x\}$ deformation retracts onto X^1 and $H_n(X - \{x\}, X^1) = 0$ for all n . The same reasoning shows the right-hand vertical map is an isomorphism. There is a similar commutative diagram with f replaced by a homeomorphism g that is homotopic to the identity and equals f in a neighborhood of x , with g the identity outside a disk in X containing x and y . Since g is homotopic to the identity, it induces the identity across the top row of the diagram, and since g equals f near x , it induces the same map as f in the bottom row of the diagram, by excision. It follows that the map f_* in the upper row is the identity.

This example generalizes to surfaces of any odd genus by adding symmetric pairs of tori at the left and right. Examples for even genus are described in one of the exercises.

Fixed point theory is a well-developed side branch of algebraic topology, but we touch upon it only occasionally in this book. For a nice introduction see [Brown 1971].

Simplicial Approximations to CW Complexes

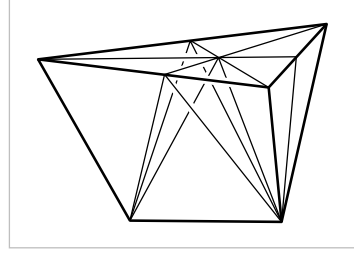
The simplicial approximation theorem allows arbitrary continuous maps to be replaced by homotopic simplicial maps in many situations, and one might wonder about the analogous question for spaces: Which spaces are homotopy equivalent to simplicial complexes? We will show this is true for the most common class of spaces in algebraic topology, CW complexes. In the Appendix the question is answered for a few other classes of spaces as well.

Theorem 2C.5. *Every CW complex X is homotopy equivalent to a simplicial complex, which can be chosen to be of the same dimension as X , finite if X is finite, and countable if X is countable.*

We will build a simplicial complex $Y \simeq X$ inductively as an increasing union of subcomplexes Y_n homotopy equivalent to the skeleta X^n . For the inductive step, assuming we have already constructed $Y_n \simeq X^n$, let e^{n+1} be an $(n+1)$ -cell of X attached by a map $\varphi: S^n \rightarrow X^n$. The map $S^n \rightarrow Y_n$ corresponding to φ under the homotopy equivalence $Y_n \simeq X^n$ is homotopic to a simplicial map $f: S^n \rightarrow Y_n$ by the simplicial approximation theorem, and it is not hard to see that the spaces $X^n \cup_{\varphi} e^{n+1}$ and $Y_n \cup_f e^{n+1}$ are homotopy equivalent, where the subscripts denote attaching e^{n+1} via φ and f , respectively; see Proposition 0.18 for a proof. We can view $Y_n \cup_f e^{n+1}$ as the mapping cone C_f , obtained from the mapping cylinder of f by collapsing the domain end to a point. If we knew that the mapping cone of a simplicial map was a simplicial complex, then by performing the same construction for all the $(n+1)$ -cells of X we would have completed the induction step. Unfortunately, and somewhat surprisingly, mapping cones and mapping cylinders are rather awkward objects in the

simplicial category. To avoid this awkwardness we will instead construct simplicial analogs of mapping cones and cylinders that have all the essential features of actual mapping cones and cylinders.

Let us first construct the simplicial analog of a mapping cylinder. For a simplicial map $f: K \rightarrow L$ this will be a simplicial complex $M(f)$ containing both L and the barycentric subdivision K' of K as subcomplexes, and such that there is a deformation retraction r_t of $M(f)$ onto L with $r_1|_{K'} = f$. The figure shows the case that f is a simplicial surjection $\Delta^2 \rightarrow \Delta^1$. The construction proceeds one simplex of K at a time, by induction on dimension. To begin, the ordinary mapping cylinder of $f: K^0 \rightarrow L$ suffices for $M(f|_{K^0})$. Assume inductively that we have already constructed $M(f|_{K^{n-1}})$. Let σ be an n -simplex of K and let $\tau = f(\sigma)$, a simplex of L of dimension n or less. By the inductive hypothesis we have already constructed $M(f: \partial\sigma \rightarrow \tau)$ with the desired properties, and we let $M(f: \sigma \rightarrow \tau)$ be the cone on $M(f: \partial\sigma \rightarrow \tau)$, as shown in the figure. The space $M(f: \partial\sigma \rightarrow \tau)$ is contractible since by induction it deformation retracts onto τ which is contractible. The cone $M(f: \sigma \rightarrow \tau)$ is of course contractible, so the inclusion of $M(f: \partial\sigma \rightarrow \tau)$ into $M(f: \sigma \rightarrow \tau)$ is a homotopy equivalence. This implies that $M(f: \sigma \rightarrow \tau)$ deformation retracts onto $M(f: \partial\sigma \rightarrow \tau)$ by Corollary 0.20, or one can give a direct argument using the fact that $M(f: \partial\sigma \rightarrow \tau)$ is contractible. By attaching $M(f: \sigma \rightarrow \tau)$ to $M(f|_{K^{n-1}})$ along $M(f: \partial\sigma \rightarrow \tau) \subset M(f|_{K^{n-1}})$ for all n -simplices σ of K we obtain $M(f|_{K^n})$ with a deformation retraction onto $M(f|_{K^{n-1}})$. Taking the union over all n yields $M(f)$ with a deformation retraction r_t onto L , the infinite concatenation of the previous deformation retractions, with the deformation retraction of $M(f|_{K^n})$ onto $M(f|_{K^{n-1}})$ performed in the t -interval $[1/2^{n+1}, 1/2^n]$. The map $r_1|_K$ may not equal f , but it is homotopic to f via the linear homotopy $tf + (1-t)r_1$, which is defined since $r_1(\sigma) \subset f(\sigma)$ for all simplices σ of K . By applying the homotopy extension property to the homotopy of r_1 that equals $tf + (1-t)r_1$ on K and the identity map on L , we can improve our deformation retraction of $M(f)$ onto L so that its restriction to K at time 1 is f .



From the simplicial analog $M(f)$ of a mapping cylinder we construct the simplicial ‘mapping cone’ $C(f)$ by attaching the ordinary cone on K' to the subcomplex $K' \subset M(f)$.

Proof of 2C.5: We will construct for each n a CW complex Z_n containing X^n as a deformation retract and also containing as a deformation retract a subcomplex Y_n that is a simplicial complex. Beginning with $Y_0 = Z_0 = X^0$, suppose inductively that we have already constructed Y_n and Z_n . Let the cells e_α^{n+1} of X be attached by maps $\varphi_\alpha: S^n \rightarrow X^n$. Using the simplicial approximation theorem, there is a homotopy from φ_α to a simplicial map $f_\alpha: S^n \rightarrow Y_n$. The CW complex $W_n = Z_n \cup_\alpha M(f_\alpha)$ contains a